HU Zixin

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EDUCATION

2020-2025: Ph.D., School of Biological Sciences, Nanyang Technological University, Singapore (cGPA: 4.88/5.0)
2016-2020: Bachelor of Science in Biological Science, School of Life Sciences, Fudan University, Shanghai, China

AWARDS

2020-2024: Tier 3 Project Scholarship
2019-2020: First Prize of the Scholarship for Outstanding Students
2017-2018: Second Prize of Fosunpharma Scholarship
2017-2018: Professional Scholarship for undergraduate students
2016-2017: Second Prize of the Scholarship for Outstanding Students

INTERESTS

Research: biomolecular modelling; molecular dynamics (MD) simulations; coarse-grained (CG) model; multiscale model; intrinsically disordered proteins (IDPs); liquid-liquid phase separation (LLPS); cytoskeleton; biomolecular assembly
Personal: reading; pop and folk music; watching videos of toucan, capybara, koala, and pika; waiting for Murakami Haruki to be awarded the Nobel Prize every October

PERSONALITY

Independent; open-minded; resilient; systematic thinker; biology & math lover

SKILLS

Software: LAMMPS, GROMACS, PyMOL, VMD, Inkscape, Blender
Programming Languages: C++, Python, MATLAB, JAVA, Linux command

RESEARCH EXPERIENCES

Postdoctoral Researcher (Kyoto U.) *Supervisors: Prof. Shoji Takada*

Project 1: *(independent, manuscript in preparation)* *Time: 2025-now*

Mesoscale study on the role of condensates in F-actin bundling

We developed a mesoscale model to investigate how phase-separated condensates affect F-actin organization and bundling. The model captures interactions between condensates and actin filaments, allowing us to examine how condensate properties and interaction strength regulate filament architecture.

Ph.D. student (NTU) *Supervisors: Prof. Lanyuan Lu & Prof. Lars Nordenskiöld*

Project 2: *(independent, published)* *Time: 2024-2025*

A Systematic Investigation on the LLPS propensity of IDPs

We designed a systematic procedure to examine the consistency of the LLPS

propensity of IDPs from simulation and experiment in a small database. We derived a recommended cutoff for multivalency giving rise to homotypical LLPS.

Project 3: *(independent, published)* *Time: 2021-2024*

Refined Bonded Terms in CG Model for IDPs Improve Backbone Conformations

We developed and implemented bonded potentials for IDPs into the coarse-grained model, Mpipi model. Results show improvement in the secondary structure propensity (SSP) of IDPs and capture the transient secondary structure without compromising the performance on the radius of gyration (Rg) and LLPS propensity.

Project 4: *(collaborative, published)* *Time: 2020-2022*

The Intrinsically Disordered Region of Coronins Fine Tunes Oligomerization and Actin Polymerization

Coronin is a class of multi-domain protein (MDP). Experiments showed that IDR tunes the oligomerization in yeast coronin but not in mice coronin. We used the hybrid CG model (Gō-like & KH model) to investigate this effect.

Research assistant (NTU) *Supervisor: Prof. Yuguang Mu*

Project 5: *(independent, finished)* *Time: July-August 2019*

Molecular Dynamic (MD) Simulation Study on Bacterial Transglycosylase

Bacterial transglycosylase is an enzyme involving in the synthesis of glycan. We used MD simulation and docking to perform virtual screening on drug database to seek potential hits for antibiotics. We ended up with several potential lead compounds.

Undergraduate student (Fudan U.) *Supervisor: Prof. Motoyuki Hattori*

Project 6: *(final year project)* *Time: 2019-2020*

Study on the Structure of the Soluble Domains of MRS2 in Plants

MRS2 is a family of magnesium ion transporters. Mutations on MRS2 are reported to induce morphological changes in *Arabidopsis thaliana* and *Oryza sativa*. We hypothesized the soluble domain to be magnesium sensor in the transportation process. We obtained purified soluble domain of MRS2 and aimed to solve the structure for a deeper understanding of it.

Project 7: *(summer project)* *Time: July-August 2018*

Study on the Structure of the Calcium Homeostasis Modulator 1 (CALHM1) Ion Channel

CALHM1 is a voltage-gated ion channel modulating calcium homeostasis, which is related to several kinds of neurological diseases and is essential in the functions of type Ⅱ bud cells. We obtained purified full-length CALHM1 from Human cell membranes and aimed to solve the structure with cryo-electronic microscope (cryo-EM) to understand the function of CALHM1.

CONFERENCES

24-26 Sept 2025: The 63rd Annual Meeting of the Biophysical Society of Japan (Nara, Japan)

14-17 May 2024: Cellular mechanisms driven by phase separation (EMBL, Germany)

1-3 Mar 2023: Phase Separation Regulated Life, In and Outside of Cells (Singapore)

PUBLICATIONS

1. **Zixin Hu**, Tiedong Sun, Lars Nordenskiöld, and Lanyuan Lu (2025) Systematic

Investigation of the Liquid-Liquid Phase Separation Propensity of Intrinsically Disordered Proteins by Molecular Simulations *The Journal of Physical Chemistry B* (https://pubs.acs.org/doi/10.1021/acs.jpcb.5c02419)

2. **Zixin Hu**, Tiedong Sun, Wenwen Chen, Lars Nordenskiöld, and Lanyuan Lu (2024) Refined Bonded Terms in Coarse-Grained Models for Intrinsically Disordered Proteins Improve Backbone Conformations *The Journal of Physical Chemistry B* Article (https://pubs.acs.org/doi/10.1021/acs.jpcb.4c02823)
3. Xiao Han[1], **Zixin Hu**[1], Wahyu Surya, Qianqian Ma, Feng Zhou, Lars Nordenskiöld, Jaume Torres, Lanyuan Lu, Yansong Miao* (2023) The Intrinsically Disordered Region of Coronins Fine Tunes Oligomerization and Actin Polymerization. *Cell Report* (https://doi.org/10.1016/j.celrep.2023.112594)